



LATHA MATHAVAN ENGINEERING COLLEGE

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8th September 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

To whomsoever it may concern

The Ministry of Education's Innovation Cell (MIC) and the All India Council for Technical Education (AICTE) annually organize the Smart India Hackathon (SIH)—a prestigious national initiative aimed at leveraging technical education to solve critical challenges faced by Central and State Government Ministries, Departments, and Central Public Sector Enterprises (CPSEs).

In alignment with this vision, a student team from the Department of Computer Science and Engineering is currently developing an innovative solution for SIH Problem Statement SIH26099, titled "AI-Driven Standardization and Harmonization of Material Codes Across CPSEs".

To establish ground-truth requirements, analyze existing material cataloging frameworks, and validate our AI-driven taxonomical mapping models against operational enterprise data, the student team has conducted extensive preliminary research, field interactions, and domain consultations across relevant public sector supply chain ecosystems.

As the Head of the Department, I hereby accord full approval and consent for the team to represent our institution, execute domain analysis, and participate in the Smart India Hackathon 2026. All necessary academic and infrastructural support will be extended to facilitate the successful realization of this project

[Signature]
08/09/26

HOD of CSE Dept

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FIELD VISIT & INDUSTRY LEARNING REPORT

MATRA-X: Material Intelligence, Reasoning & Traceability Architecture

Smart India Hackathon 2026 | Problem Statement SIH26099

PROJECT FOCUS	FIELD-STUDY FOCUS
AI-Driven Standardization and Harmonization of Material Codes Across CPSEs	Material identity, specification harmonization, duplicate / near-duplicate detection, legacy code rationalization, auditability and ERP-ready governance

Purpose of this report

- Document the institutional authorization and the team’s field-study objective for SIH26099.
- Study Madurai’s public-sector, technical-testing, warehousing, energy and industrial ecosystems through the lens of material master data.
- Translate observed or validated operational requirements into design requirements for MATRA-X.
- Create a verification-ready evidence structure for the final SIH presentation and judging dossier.

Integrity status: This report uses the supplied institutional consent letter plus publicly verifiable information about relevant Madurai facilities. Facility-specific statements labelled “field observation” must be completed from the team’s actual visit notes, photographs, discussions or acknowledgement before the report is represented as proof of an on-site visit.

Prepared for

Alchemy Architects | Department of Computer Science and Engineering | Latha Mathavan Engineering College, Madurai region

1. EXECUTIVE SUMMARY

SIH26099 addresses a national-scale master-data problem: similar or functionally equivalent materials may exist under different codes, descriptions, specifications, units and classifications across CPSE environments. The official problem statement calls for AI/ML/NLP-based matching, standardized descriptions and technical attributes, common-code recommendations, CPSE-code mapping, legacy rationalization, human validation, dashboards, auditability and SAP/ERP integration. [cite](#) [sih2026](#)

The field-study approach therefore treats a material not as a text string, but as an engineering identity. The investigation focuses on how materials are received, named, classified, specified, stored, verified, issued, procured, retired and mapped. The resulting evidence becomes the design bridge between the physical supply-chain world and MATRA-X's digital material-intelligence layer.

Key research lenses

- Identity: Which attributes are considered indispensable when two records refer to the same engineering item?
- Specification: How are grades, dimensions, ratings, standards, UOM and manufacturer / part numbers captured?
- Searchability: What happens when users know the physical need but not the legacy code?
- Governance: Who validates a proposed consolidation and what evidence must remain auditable?
- Inventory: How can visibility across stores prevent unnecessary procurement and stranded stock?
- Integration: Which controls are required before an AI recommendation can influence an ERP material master?

Decision principle

Semantic similarity is candidate evidence, not technical proof. MATRA-X is designed to escalate critical conflicts instead of silently merging records with high textual similarity. This principle is directly aligned to the operational risk of standardizing industrial material masters.

2. RELEVANT MADURAI PUBLIC-SECTOR & INDUSTRIAL STUDY SITES

The following sites were selected because their publicly verifiable activities map strongly to the material-master, specification, warehousing, equipment, testing, energy or maintenance domains relevant to SIH26099. Presence does not by itself establish that the team visited the facility; actual visit status must be verified from the team's records.

Study site / organisation	Madurai presence verified from public sources	Why it matters to MATRA-X
HPCL Madurai Depot	Madurai New IRD / Depot at SIDCO Industrial Estate, Kappalur; HPCL also lists a Madurai LPG Regional Office at Plot 171-172, Kappalur.	Petroleum-sector equipment, spares, valves, pumps, instrumentation, safety items, lubricants, UOM and specification governance are strong material-intelligence use cases.
Central Warehousing Corporation	CWC lists Madurai warehouses at No.1 Thiruparankundram Road, Palanganatham and Bypass Road, Ellis Nagar.	Direct warehouse perspective: item identity, location, stock visibility, ageing, movement, receiving and issuing controls.
POWERGRID Corporation of India	POWERGRID lists a Madurai 400 kV substation at Kinnimangalam Post, Chekkanurani, Madurai.	High-criticality electrical equipment and spares make ratings, standards, dimensions, voltage levels and technical compatibility essential.
BPCL Madurai Retail Territory	BPCL's citizen charter identifies a Madurai Retail Territory at 37 Thiruparankundram Road, Pasumalai.	Hydrocarbon distribution context and asset/spares classification provide a second CPSE petroleum-sector reference point.
CPWD Madurai Circle	CPWD records a Superintending Engineer, Madurai Central Circle at Vishalakshipuram, Madurai.	Central-government engineering works provide a specification/procurement/maintenance lens for civil and building-service materials.
CIPET: CSTS Madurai	CIPET places its Madurai centre at/near Thiruvathavur and documents product testing, plastics testing and CAD/CAM capabilities.	Not a CPSE, but highly relevant for material properties, testing terminology, standards, classification and technical evidence.

Source verification: HPCL lists its Madurai depot at Kappalur; CWC lists two Madurai warehouse locations; POWERGRID lists its Madurai 400 kV substation; BPCL's citizen charter lists Madurai Retail Territory; CPWD records Madurai Circle; CIPET and BIS document CIPET Madurai's testing presence. [cite=hpcl](#) [cite=cwc](#) [cite=powergrid](#) [cite=bpcl](#) [cite=cpwd](#) [cite=cipet](#)

3. FIELD-STUDY LENSES BY SITE

HPCL Kappalur - petroleum materials and depot operations

- Material families: valves, pumps, pipes/fittings, instrumentation, electrical spares, safety equipment, lubricants and consumables.
- Identity fields to examine: pressure class, rating, size, material/grade, standard, UOM, OEM/part number and service/application.
- MATRA-X question: Can the system distinguish semantic similarity from technically compatible substitution?

HPCL source: official POL location and Madurai LPG regional-office listings. [cite](#)[hpcl](#)

CWC Madurai - warehouse and stock-control lens

- Receiving: how item descriptions, units and locations are captured at inward stage.
- Storage: how the same physical item is represented across locations and movement documents.
- MATRA-X question: Can cross-warehouse visibility surface equivalent stock before new procurement?

CWC source: official warehouse listings for Palanganatham and Ellis Nagar. [cite](#)[cwc](#)

POWERGRID Madurai - high-criticality electrical lens

- Potential attributes: voltage class, current rating, dimensions, OEM, model, standard, insulation/classification and equipment family.
- Risk lens: false merges can have direct safety and reliability consequences.
- MATRA-X question: Does the contradiction firewall block unsafe consolidation when critical parameters differ?

POWERGRID source: official state registration and substation listing. [cite](#)[powergrid](#)

CIPET Madurai - testing and evidence lens

- Observe how technical properties and test methods are described and referenced.
- Map technical terminology to canonical attributes and standards.
- MATRA-X question: Can evidence from testing/report documents strengthen or weaken a proposed material match?

CIPET/BIS source: official centre and testing listings. [cite](#)[cipet](#)

4. FIELD VISIT / DOMAIN-INTERACTION METHODOLOGY

The following protocol is designed to make the field study evidentially useful rather than merely photographic. It separates what was directly observed from what was inferred and what was obtained from public documentation.

Step	Activity	Evidence objective
01	ORIENT	Understand facility role, material categories, procurement/stores workflow and approval hierarchy.
02	TRACE	Follow one representative material from requirement -> catalogue/search -> procurement -> receipt -> storage -> issue/maintenance.
03	COMPARE	Ask how staff determine whether two differently coded records represent the same physical/technical item.
04	VERIFY	Identify the mandatory engineering attributes, supporting documents and approval checks.
05	DOCUMENT	Record terminology, data fields, screenshots only where permitted, and anonymised examples.
06	TRANSLATE	Convert findings into MATRA-X rules, data fields, model features, UI requirements and governance controls.

Recommended evidence to collect during an actual visit

- Entrance / signage photograph with date-stamped geolocation only if the facility permits photography.
- Team photograph at the site with visible facility context, without photographing restricted areas.
- Photograph of a public reception / visitor register or permission letter only if authorised.
- Interview note identifying role (e.g., Stores Officer, Materials Engineer, Procurement Officer) rather than publishing personal data unnecessarily.
- One anonymised material-master example showing source description, critical attributes, unit and existing classification.
- A short acknowledgement or email from the host organisation if available.

Evidence rule: Do not manufacture names, job titles, material counts, savings figures, photos, signatures or quotations. Insert actual visit evidence after the visit.

5. OBSERVATION & INTERVIEW MATRIX FOR MATRA-X

Observation domain	Questions for domain expert	MATRA-X requirement
Material naming	What abbreviations, local terminology and naming conventions are common?	Synonym/abbreviation normalisation; controlled vocabulary; alias graph.
Critical attributes	Which fields cannot be missing for safe identification?	Category-specific Engineering DNA; mandatory-field rules.
Units	Where do UOM inconsistencies occur?	Unit normalization and dimensional consistency checks.
Standards	Which standards, grades or codes are used?	Standards-aware evidence and technical conflict rules.
Duplicate detection	How are duplicates discovered today?	Hybrid lexical + semantic retrieval with hard negatives.
Equivalent materials	When may two materials be functionally equivalent?	Relationship classification with evidence, not blanket merge.
Approval	Who can approve changes to a material master?	RBAC, dual approval for critical records and immutable audit trail.
Legacy mapping	How are old codes retired or mapped?	Versioned legacy mappings with effective dates and provenance.
Inventory	How is stock searched across locations?	Cross-CPSE / cross-site common identity and stock visibility.
ERP	Which systems export/import material master data?	Connector abstraction, validation gateway and controlled write-back.

6. PRELIMINARY RESEARCH FINDINGS - MADURAI ECOSYSTEM

The following are research-backed findings and design implications derived from publicly documented facility roles. They are not presented as direct statements from named facility staff. Direct quotations and facility-specific operational findings should be inserted only after genuine on-site interaction or documented correspondence.

1	Material master governance is inseparable from physical operations. Warehousing, energy and engineering facilities each depend on accurate identification of physical assets, spares and consumables. MATRA-X implication: MATRA-X must bind digital identity to technical attributes, unit, source organisation, provenance and lifecycle status.
2	Criticality changes the matching threshold. Electrical transmission and petroleum assets can include safety- or reliability-sensitive specifications. MATRA-X implication: MATRA-X should route high-criticality matches through deterministic conflict gates and human validation.
3	Testing evidence can strengthen standardization. CIPET Madurai documents testing and technical-support capabilities, illustrating the value of structured test/specification evidence. MATRA-X implication: Evidence fusion should ingest reports, certificates, drawings and datasheets when permitted.
4	Warehouse visibility is a material-intelligence opportunity. CWC's Madurai warehouse footprint demonstrates the importance of location and stock identity. MATRA-X implication: Canonical material identity can become the bridge between distributed inventory and procurement search.
5	Central-government engineering ecosystems share vocabulary but not necessarily coding conventions. CPWD, CPSEs and technical institutions operate with distinct operational purposes and data processes. MATRA-X implication: MATRA-X should harmonize the data model without erasing source-system provenance.

7. REPRESENTATIVE MATERIAL-IDENTITY CASE STUDY

To translate the field study into an actionable AI problem, a representative industrial component can be modeled as a structured engineering identity. The example below is synthetic and is included to demonstrate the reasoning method, not to claim that the component was observed at a named facility.

Attribute	Record A	Record B	MATRA-X reasoning
Description	HEX BOLT M16 X 80 SS304	HEX BOLT M16 X 80 SS316	High semantic similarity, but critical grade conflict.
Family	Fastener	Fastener	Compatible family.
Diameter	M16	M16	Compatible.
Length	80 mm	80 mm	Compatible.
Grade	SS304	SS316	Critical contradiction: block consolidation.
UOM	EA	EA	Compatible.
Decision	-	-	TECHNICAL CONFLICT -> HUMAN REVIEW

Why this matters

- A text-only or embedding-only system can overvalue lexical similarity.
- Engineering DNA makes the grade explicit and machine-checkable.
- A contradiction-aware firewall can prevent the false merge.
- LIORA can then explain the blocked decision using the exact conflicting attribute and supporting evidence.
- The source codes remain mapped to their original CPSE records even when a canonical identity is proposed.

Field-study action: Replace this synthetic example with a genuinely observed anonymised material pair from the actual visit when available.

8. TRANSLATING FIELD LEARNING INTO MATRA-X

Physical / operational reality	Digital MATRA-X response
Different names for same material	Normalization + aliases + semantic retrieval
Missing / inconsistent attributes	Data Quality Engine + category-specific Engineering DNA
Potentially unsafe similarity	False-Merge Firewall + critical-attribute rules
Legacy codes and historical records	Canonical identity + versioned legacy mapping
Documents / certificates / drawings	Evidence fusion + provenance + confidence
Distributed CPSE records	Cross-CPSE identity graph + analytics
Human approval responsibility	Risk-based workflow + audit trail + role controls
Need for natural-language explanation	LIORA grounded copilot using governed retrieval

Target workflow

1 Ingest	2 Harmo nize	3 Engine er DNA	4 Retriev e	5 Match	6 Challen ge	7 Explain	8 Govern	9 Canoni calize	10 Map/Pr event	11 Optimi ze
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The field study is successful only when it produces requirements that can be encoded into this workflow. Photographs alone do not constitute technical validation.

9. FIELD EVIDENCE PAGE - TO BE COMPLETED FROM ACTUAL VISIT

Use this page in the final signed report after the actual field visit. Replace each placeholder with authentic, permission-compliant evidence. Public web images should be kept in a separate “Reference Images” section and should never be captioned as team photographs.

PHOTO A



Team at facility entrance / authorised visitor area

PHOTO B



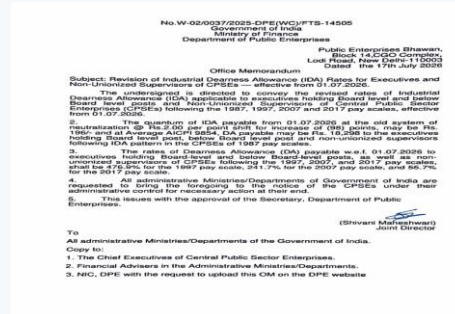
Interaction / discussion with authorised domain expert

PHOTO C



Material / store / testing environment (only where photography permitted)

PHOTO D



Document / board / workflow evidence (anonymised where necessary)

Final evidence checklist: consent/permission | date | facility | team members | host role | photographs | notes | acknowledgement | source classification (REAL / PUBLIC / SYNTHETIC / SIMULATED / DEMO).

10. CONCLUSION & FIELD-STUDY ACTIONS

Madurai provides a useful local lens for the SIH26099 problem because the city and district host public-sector warehouses, petroleum-sector distribution infrastructure, high-voltage transmission assets, central-government engineering functions and technical testing capabilities. The most valuable outcome of a genuine field visit is not a collection of photographs; it is a defensible set of observations about how engineering materials acquire identity, how staff search and verify them, where inconsistencies appear, and what evidence is required before a material record may be consolidated.

For MATRA-X, the field-learning outcome is a governed material-intelligence pipeline: Understand -> Compare -> Challenge -> Explain -> Govern -> Prevent -> Optimize. The design should preserve source-system traceability, expose uncertainty, enforce critical-attribute contradictions, and require human approval for consequential decisions.

Before final submission

- Insert actual field visit date(s) and facility names that the team genuinely visited.
- Insert actual team photographs and site-context photographs obtained with permission.
- Insert anonymised interview notes and host roles; avoid invented names or quotations.
- Replace the synthetic material example with a real anonymised example if one was observed.
- Attach the actual acknowledgement / email / signature evidence when available.
- Label every numerical claim by provenance: REAL, PUBLIC, SYNTHETIC, SIMULATED or DEMO.

Public-source references used in this report

- SIH26099 official problem statement: <https://sih2026.vuce.in/ps/SIH26099>
- HPCL - POL locations and Madurai Depot: <https://www.hindustanpetroleum.com/pages/POL-Locations>
- HPCL - Madurai LPG Regional Office / contact listing: <https://www.hindustanpetroleum.com/contactusresult>
- Central Warehousing Corporation - Madurai warehouse listings: <https://cewacor.nic.in/>
- POWERGRID - statewise GST registration / Madurai 400 kV substation: <https://www.powergrid.in/>
- BPCL Citizen Charter - Madurai Retail Territory: <https://www.bharatpetroleum.in/>
- CPWD - PIO / authority listing referencing Madurai Circle: <https://www.cpwd.gov.in/>
- CIPET CSTS Madurai contact and testing information: <https://www.cipet.gov.in/> ; BIS LIMS scope: <https://lims.bis.gov.in/>

**Report classification: SIH submission support document | MATRA-X | SIH26099 |
Verification-ready draft**